Name (please print): _____

Nmap Exercise 1 ~ 5: ***(You may use Nmap or its GUI version Zenmap for these exercises)***

If not already installed on your system, download the most current version of Nmap from https://nmap.org/download.html.

Submission Requirements: type your answers and paste screen shots directly in this document for submission.

Nmap ("Network Mapper") is a free and open source utility for network exploration and/or security auditing. Many systems and network administrators also find it useful for tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime.

Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

It was designed to rapidly scan large networks, but works fine against single hosts. Nmap runs on all major computer operating systems, and both console and graphical versions are available.

Nmap employs the various active stack fingerprinting techniques mentioned in your slides (except for the fragmentation handling and ICMP error message queuing) by using the -O option.

Remember, the accuracy of the determination of the remote OS is largely dependent on at least one open port on the target system.

In the next few exercises, you will be using Nmap to identify the OS running on the ***target*** (Your own) ***system***(s). [Only scan systems on your own personal network***. You are only authorized to scan the system(s)/network in which you ow or have permission to scan.***

Exercise 1: Active stack fingerprinting option:

- From a Command Line Interface (CLI), type the following (at the command prompt only type what's in **bold**, on one line):

  **Notes:*

  you must be in the directory that Nmap is currently installed

  <target_IP_address> is to be replaced by your own IP address of your system

  nmap -O -v <target_IP_address>

  Syntax breakdown:

  nmap: program name

  -O: this is the letter O - not the number zero; program option for active stack fingerprinting of remote OS

-v: **verbose mode**

<target_IP_address>: IP address of the target system

- Examine your results and answer the following questions:
  - What OS did Nmap determine is running on the target system? Is it correct? ***(Make sure to screenshot the lower command line starting with "Read...")***

Windows

```
IP ID Sequence Generation: All zeros

Initiating SYN Stealth Scan at 15:35
Scanning DESKTOP-J7PJJ49 (192.168.1.155) [1000 ports]
Discovered open port 445/tcp on 192.168.1.155
Discovered open port 139/tcp on 192.168.1.155
Discovered open port 135/tcp on 192.168.1.155
Completed SYN Stealth Scan at 15:35, 0.15s elapsed (1000 total ports)
Initiating OS detection (try #1) against DESKTOP-J7PJJ49 (192.168.1.155)
Nmap scan report for DESKTOP-J7PJJ49 (192.168.1.155)
Host is up (0.00098s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1809 - 21H2
Network Distance: 0 hops
TCP Sequence Prediction: Difficulty=263 (Good luck!)
IP ID Sequence Generation: Incremental

Read data files from: C:\Program Files (x86)\Nmap
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (10 hosts up) scanned in 66.15 seconds
           Raw packets sent: 11967 (538.620KB) | Rcvd: 10716 (452.832KB)
```

Exercise 2: in this exercise, you will combine TCP and UDP port scans with application version detection and OS fingerprinting, in one command:

- The syntax to perform this is (at the command prompt only type what's in **bold**, on one line):

  nmap -A -sS -sU -v <target_IP_address>/24

Syntax breakdown:

nmap: program name

-A: program option combining application vendor/version detection on open ports and active stack fingerprinting of remote OS

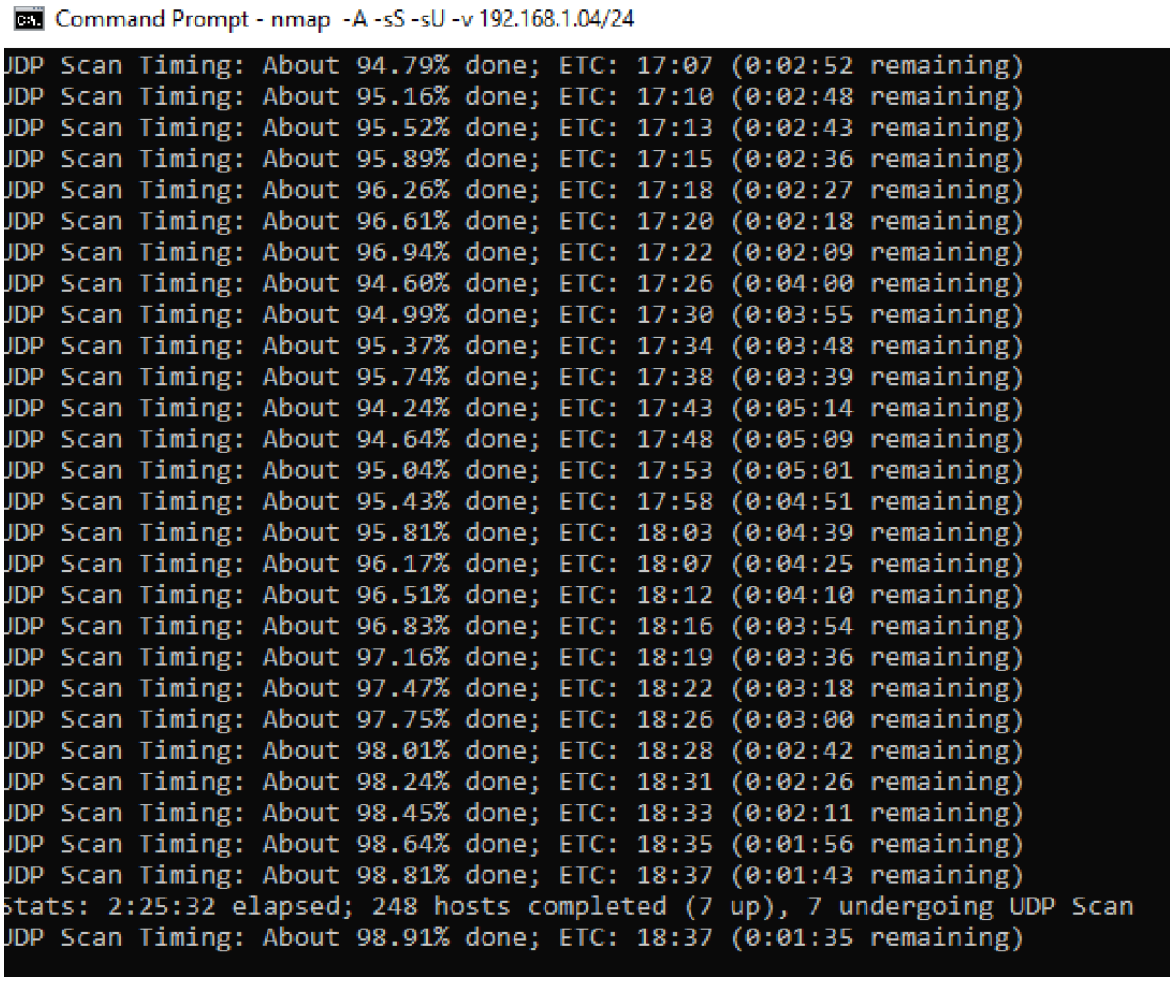
-sS: program option for TCP SYN scan

-sU: program option for UDP port scan

-v: **verbose mode**

<target_IP_address>: the IP address of the target system

- Examine your results:
- Record the application vendor(s) and version(s) running on the various UDP and TCP ports: *This may be typed out or a screen shot*



Exercise 3: in this exercise, it is worth noting that Nmap does offer ping sweep capabilities with the -sP option:

- From a Command prompt, type the following (at the command prompt only type what's in **bold**, on one line)::

  <class_IP_range> is to be replaced by your own IP address of your personal network

C: **nmap -sP -v <class_IP_range>/24**

Syntax breakdown:

nmap: program name

-sP: program option for ping sweep

-v: verbose mode

class_IP_range/24: replace class_IP_range with the network block to scan (e.g., 10.10.10.0/24)

- Examine your results:
- In the output, look for the text "**Host is up**" and record these systems' IP addresses here:

192.168.1.100 (Living Room)

192.168.1.152 (RokuTV)

192.168.1.160 (Someone's phone)

192.168.1.158 (Someone's phone)

```
Initiating Parallel DNS resolution of 1 host. at 15:55
Completed Parallel DNS resolution of 1 host. at 15:55, 0.01s elapsed
Nmap scan report for DESKTOP-J7PJJ49 (192.168.1.155)
Host is up.
Read data files from: C:\Program Files (x86)\Nmap
Nmap done: 256 IP addresses (9 hosts up) scanned in 5.82 seconds
           Raw packets sent: 511 (14.308KB) | Rcvd: 14 (392B)

C:\Users\Briana Laws>
```

Exercise 4: Service Vendor and Version Detection using nmap:

- The syntax to detect the vendor and version of a service running on a TCP port is (at the command prompt only type what's in **bold**, on one line):

  nmap -sV -vv <target_IP_address>

Syntax breakdown: ***[scan 192.168.1.152]***

nmap: program name

-sV: program option for service vendor and version

-vv: program option for double verbose output

<target_IP_address>: the IP address of the target system

- Examine your results:
- Record your results: *This may be typed out or a screen shot*

PORT *STATE* *SERVICE* *VERSION (***Screenshot)**

--- --- --- ---

```
PORT    STATE SERVICE      REASON          VERSION
135/tcp open  msrpc        syn-ack ttl 128 Microsoft Windows RPC
139/tcp open  netbios-ssn  syn-ack ttl 128 Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds? syn-ack ttl 128
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Read data files from: C:\Program Files (x86)\Nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (9 hosts up) scanned in 219.94 seconds
          Raw packets sent: 9683 (417.940KB) | Rcvd: 10228 (413.312KB)

C:\Users\Briana Laws>
```

Exercise 5: TCP SYN scan:

Let's build on this common port scan so that your actions are *stealthier*.

In the next example, you will add two additional options: the -g and the –p options:

The *-g option* specifies the source port on the scanning machine (your system).

The most common option to use here is port 80 (because it's allowed through most

border network devices). The *-p option* sets the ports on the target system to scan.

- Using the -sS flag option, you will send the target system a SYN packet, but upon receiving the SYN/ACK from the target system, your system will not respond with an ACK packet (thus, completing the 3-way handshake). Instead, your system will respond with a RST packet (the port is then considered open). If you receive an RST/ACK packet from the target system, it indicates that there is not a running process on that port, or the process running on the port is not listening for connections.
- The syntax to perform a TCP SYN scan is (at the command prompt only type what's in **bold**, on one line):

  nmap -sS -vv -g 80 -p 80,88,135,139,389,445 <target_IP_address>

Syntax breakdown:

nmap: program name

-sS: program option for TCP SYN scan

-vv: program option for double verbose output

-g 80: program option that specifies the source port on the scanning machine (your system)

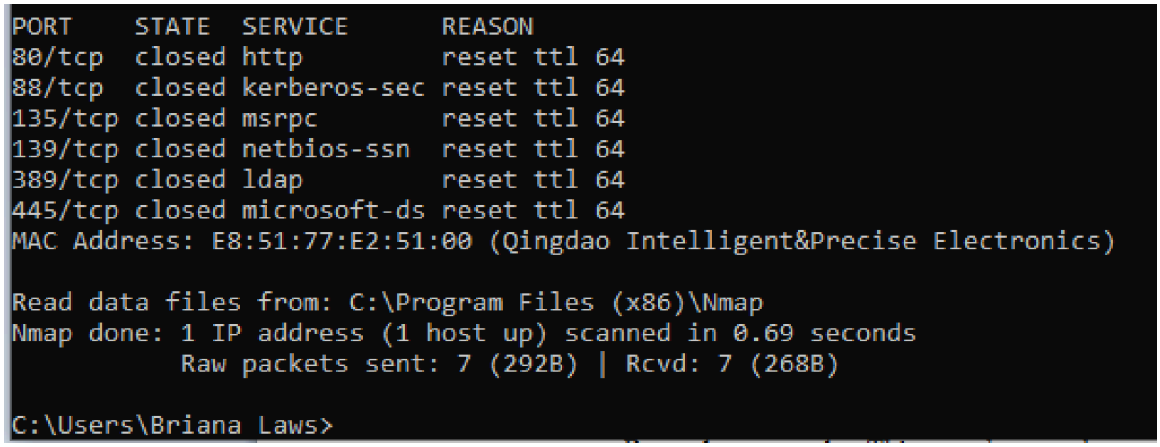
-p 80,88,135,139,389,445: specifies the ports on the target system to scan

<target_IP_address>: the IP address of the target system ***(scanning one network node)***

- Examine your results:
- Record your results: *This may be typed out or a screen shot*

PORT *STATE* *SERVICE*

--- --- ---



- Repeat step #2, this time scanning for all open port numbers.
- Record your results:

PORT *STATE* *SERVICE*

--- --- ---

```
PORT    STATE  SERVICE      REASON
80/tcp  closed http         reset ttl 128
88/tcp  closed kerberos-sec reset ttl 128
135/tcp open   msrpc        syn-ack ttl 128
139/tcp open   netbios-ssn  syn-ack ttl 128
389/tcp closed ldap         reset ttl 128
445/tcp open   microsoft-ds syn-ack ttl 128

Read data files from: C:\Program Files (x86)\Nmap
Nmap done: 256 IP addresses (8 hosts up) scanned in 6.74 seconds
           Raw packets sent: 557 (16.364KB) | Rcvd: 68 (2.632KB)

C:\Users\Briana Laws>
```